

AI FarmTwin — Complete Website Working Explanation

“Predict. Simulate. Protect.”

AI FarmTwin is a software-based digital twin platform for farms. It creates a digital representation of a farmer's farm where the user can configure zones, plants, crops and environmental conditions, enter sensor-like values manually, monitor crop health, simulate events, perform What-If analysis, and generate reports.

The current prototype does not require physical sensors. Instead, sensor values are manually entered by the user to demonstrate how the system would behave when real sensor data is available in the future.

1. User Opens the Website

When the user opens AI FarmTwin, they first see the **authentication interface**.

The user has three main options:

- **Register**
- **Login**
- **Reset Password**

The system uses **Firebase Authentication** to securely manage user accounts.

The website does not use a common account or shared farm.

Every registered user gets their own account and their own farm data.

2. Registration

A new user selects **Register**.

They provide information such as:

- Full Name
- Farm Name
- Email
- Password
- Confirm Password

When the user submits the form, the website communicates with **Firebase Authentication**.

Firebase creates a real user account and assigns a unique **UID** to that user.

For example:

User A

Email: farmerA@gmail.com

Firebase UID: UID_A

Another user could have:

User B

Email: farmerB@gmail.com

Firebase UID: UID_B

These two users are completely independent.

3. First Login / New User Setup

After verification, the user logs in.

Firebase authenticates the user and provides the user's unique UID.

The application then checks Firestore for that UID.

If this is a completely new user, the application creates a **fresh FarmTwin environment**.

The user does not receive another user's farm.

Initially, values that have not been configured are shown as:

Awaiting Input

or

Not Entered

There are no randomly generated sensor readings or fake farm values.

4. User Enters the AI FarmTwin Dashboard

After successful login, the user enters the main AI FarmTwin application.

The main navigation contains:

1. Dashboard
2. Digital Twin
3. Crop Health
4. Plants
5. Event Detection
6. What-If Simulator
7. Reports
8. Settings
9. Logout

The entire application operates using the same centralized farm data.

5. Farm Configuration

The first important step is configuring the farm.

The user can enter information such as:

- Farm Name
- Farm Manager
- Number of Acres
- Number of Zones
- Number of Plants
- Crop information

The system is completely dynamic.

For example, if the farmer creates:

3 Zones

the system generates:

Zone A

Zone B

Zone C

If the farmer creates:

8 Zones

the system automatically creates:

Zone A

Zone B

Zone C

Zone D

Zone E

Zone F

Zone G

Zone H

There is no fixed limit such as only three zones in the application design.

6. Each Zone Gets Four Sections

Every zone is automatically divided into four sections:

North

South

East

West

For example:

ZONE A

├— North

├— South

├— East

└— West

The same structure is automatically created for every zone.

Therefore, if the farm has 10 zones, the system automatically manages:

10 zones × 4 sections = 40 sections.

Each section can have its own plant allocation, sensor information, health condition and risk information.

7. Plant Records Are Created

Based on the configured plant count, the system creates actual plant records.

For example, if a farm has:

100 plants

the system maintains 100 plant records.

These plants are distributed through the farm's zones and sections according to the application's allocation logic.

Each plant remains an individual data record.

The system does not create fake decorative plants just for visualization.

8. Plant Grouping

The physical location system groups plants into sets of **10 plants**.

For example:

Plant 1–10

→ Group N-01

Plant 11–20

→ Group N-02

Plant 21–30

→ Group N-03

The group ID depends on the section.

For example:

North → N-01

South → S-01

East → E-01

West → W-01

Each group of plants has **one shared physical address**.

So instead of giving every plant a separate physical address, the application represents the physical area containing those plants.

For example:

Zone A / North / Group N-01

If the final group contains fewer than 10 plants, the system keeps only the actual plants.

For example:

255 plants

25 groups × 10 = 250

Final group = 5 plants

No fake plants are added.

9. Dashboard

The Dashboard provides a quick overview of the farm.

It can display information such as:

- Farm name
- Farm manager

- Total acres
- Total zones
- Total plants
- Overall temperature
- Overall humidity
- Overall soil moisture
- Weather
- Overall health
- Risk summary
- Crop distribution
- Important alerts
- Charts

The Dashboard is intended to give the farmer a quick understanding of the farm without requiring them to inspect every section.

10. Manual Sensor Input

This is an important part of the current prototype.

The system currently does **not** have physical sensors connected.

Instead, the farmer/user manually enters the values that would normally come from sensors.

For example:

Temperature: 31°C

Humidity: 68%

Soil Moisture: 42%

These values are entered through the application's sensor input interface.

The important architecture is:

USER INPUT



CALCULATION ENGINE



FARM DATA



DIGITAL TWIN



HEALTH / RISK VISUALIZATION

The application does **not** randomly generate sensor values.

If a value has not been entered, it shows:

Awaiting Input

11. Section-Level Sensor Data

Each farm section can have its own environmental information.

For example:

Zone A

North

Temperature: 29°C

Humidity: 68%

Soil Moisture: 52%

The user can enter different values for another section.

For example:

Zone A

South

Temperature: 34°C

Humidity: 45%

Soil Moisture: 22%

The application treats these as separate section conditions.

This allows the Digital Twin to represent different conditions in different parts of the farm.

12. Health Calculation

Once sensor values are entered, the application processes them through the **health calculation engine**.

The system uses defined rules and thresholds to calculate the condition.

The result can be represented using statuses such as:

- Healthy
- Warning
- High Risk
- Critical
- Awaiting Input

The important concept is that the health result is **calculated from the provided values**.

For example:

Sensor Values

↓

Temperature

Humidity

Soil Moisture

↓

Health Calculation

↓

Health %

↓

Status

If the user changes a value, the result can change accordingly.

13. Digital Twin

The **Digital Twin** is the main visual Centerpiece of AI FarmTwin.

It provides a digital representation of the physical farm.

The farm is represented using a **top-down layout**.

Each zone is a connected farm plot containing:

NORTH | EAST

-----+-----

WEST | SOUTH

The sections are not unrelated cards.

They form one connected representation of the farm.

The Digital Twin displays:

- Zones
 - Sections
 - Plant positions
 - Crop information
 - Plant groups
 - Section health
 - Environmental information
-

14. Plant Visualization in the Digital Twin

The individual plants are represented using small dots.

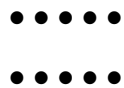
These dots correspond to the **actual plant records**.

The application does not simply draw random dots.

If the farm has 100 actual plants, the visualization represents those 100 plants.

The plants are organized in straight rows and columns.

The 10-plant group arrangement uses:



This represents 10 plants.

Group boundaries are represented using subtle dotted/dashed boundaries.

The major section boundaries are represented with clear solid lines.

15. Clicking a Plant Group

The Digital Twin does not permanently clutter the map with addresses and group information.

Instead, when the user interacts with a group, the application can show its details.

For example:

Group: N-01

Zone: A

Section: North

Plants: 10

Shared Address:

Zone A / North / Group N-01

The user can then inspect the plants belonging to that group.

16. Crop Health Page

The **Crop Health** page provides a more detailed analysis.

The user can compare different:

- Zones
- Sections
- Crops

The page can show:

- Health percentage
- Health status
- Risk level
- Temperature
- Humidity
- Soil moisture
- Plant count
- Charts

This helps the farmer identify which areas of the farm are performing well and which areas need attention.

17. Plants Page

The Plants section provides access to the plant inventory.

The main inventory is organized around **plant groups**.

For example:

Group N-01

10 plants

Zone A / North

Group N-02

10 plants

Zone A / North

The user can open a group to inspect individual plants.

The application supports searching and filtering using information such as:

- Plant ID
- Group ID
- Shared address
- Zone
- Section
- Health status

The same centralized plant records are used throughout the application.

18. Event Detection

Event Detection is separate from the normal Digital Twin.

Its purpose is to represent possible:

- Animals/wildlife
- Humans
- Vehicles

inside the farm.

Because the current prototype does not have physical cameras, these events are **simulated manually**.

The application does not pretend that a real camera is connected.

19. Event Location

When an event is created, it has an actual position on the farm map.

For example:

Animal

Zone A

North

X = 55%

Y = 40%

The marker appears at that position.

The application then determines which plant/group is closest to the event.

For example:

Event



Nearest plant



Plant's group



Group address

If the event is near Plant P006 and P006 belongs to:

Zone A / North / Group N-01

the user sees:

Current Detected Location:
Zone A / North / Group N-01

rather than an individual plant address.

This makes the location representation consistent with the farm's physical grouping system.

20. Event Movement

Events can have changing positions.

If an event moves from one group to another, the system recalculates the current group.

For example:

Initial:

Zone A / North / Group N-01

Event moves



Zone A / North / Group N-02

The displayed detected location changes automatically.

This gives Event Detection a more realistic digital-twin behavior.

21. Intrusion History

The application maintains event/intrusion information.

Each event can contain information such as:

- Event type
- Zone
- Section
- Current group
- Shared address
- Coordinates
- Entry point
- Time
- Status

The history uses the **shared group address**, not an individual plant's physical address.

22. What-If Simulator

The **What-If Simulator** allows the user to experiment with hypothetical conditions.

For example, the current condition could be:

Soil Moisture = 25%

The user can create a simulation:

Soil Moisture = 45%

The system runs the same health calculation using the simulated value.

It then compares:

CURRENT

vs

SIMULATED

The user can see changes in:

- Health
- Risk
- Status
- Warnings

This allows the farmer to explore possible outcomes before changing actual conditions.

23. Apply Simulation

The simulator can temporarily apply a simulation to the live Digital Twin.

When this happens, the application clearly indicates:

SIMULATION ACTIVE

The simulated values temporarily affect the visualization.

However, the simulation does not permanently overwrite the user's saved sensor data.

When the user selects:

Reset Simulation

the application returns to the saved live farm state.

24. Reports

The Reports page generates a summary of the user's current farm.

The report can contain:

- Farm name
- Farm manager
- Acres
- Number of zones
- Total plants
- Healthy plants
- Warning plants
- High-risk plants
- Critical plants
- Zone summaries
- Section summaries
- Risk information
- Warning information
- Plant/group information

Because the report uses the centralized farm data, it changes automatically when the farm changes.

The application can also generate a real PDF report.

25. Settings

The Settings page allows the user to modify their farm.

The user can change:

- Farm name
- Farm manager
- Acres
- Number of plants
- Number of zones
- Plant allocation
- Crop names
- Theme
- Logo

If the number of zones changes, the application automatically regenerates the required zone configuration.

If plant numbers change, the plant records and plant groups are recalculated.

These changes propagate throughout the application.

26. Centralized Data

One of the most important architectural features is that **all modules use the same farm data**.

For example:

Settings



Farm Configuration



Plant Records



Plant Groups



Digital Twin



Crop Health



Plants



Event Detection



Reports

There is not one dataset for the Dashboard and another dataset for the Plants page.

Everything is connected.

If the user changes something, the relevant parts of the application update automatically.

27. Firebase Firestore Persistence

The user's farm information is stored in **Cloud Firestore**.

The key concept is:

Firebase Authentication UID



User's Firestore data

For example:

users/

UID_A/

farmData

plants

events

settings

Another user has a different UID:

users/

UID_B/

farmData

plants

events

settings

Therefore, each user's farm remains separate.

The uploaded project history confirms that the application was designed around UID-specific Firestore storage rather than a shared farm object.

28. Privacy Between Users

Suppose User A creates:

Farm A

1000 plants

Rice

User B creates:

Farm B

500 plants

Tomato

When User A logs in:

Firebase → UID_A



Firestore → Farm A

When User B logs in:

Firebase → UID_B



Firestore → Farm B

User A cannot see User B's farm.

User B cannot see User A's farm.

Firestore security rules enforce this separation at the database level.

29. Logout

When the user clicks **Logout**:

1. Firebase signs the user out.
2. The active user becomes null.
3. Current farm data is cleared from application memory.
4. Temporary event state is cleared.
5. The application returns to the login screen.
6. The previous farm must not remain visible.

This is important because another person could then log into the same browser.

The next account starts by loading **its own UID-specific Firestore data**.

The existing implementation specifically includes resetting the in-memory farm state and events during logout.

30. Returning User

Suppose User A logs out today and comes back tomorrow.

They log in using the same email and password.

Firebase identifies the same UID:

User A

↓

UID_A

↓

Firestore

↓

Previously saved Farm A

The application restores the user's:

- Farm configuration
- Zones
- Plants
- Groups
- Sensor values
- Health information
- Events
- Settings
- Theme
- Logo

The user can continue from where they left off.

31. Password Reset

If the user forgets their password, they select:

Reset Password

They enter their registered email address.

The user follows the secure Firebase process to create a new password.

The application does not store passwords itself.

32. Complete Working Flow

The entire AI FarmTwin system can be summarized as:

OPEN WEBSITE



REGISTER / LOGIN



REGISTER



Firebase Authentication



Unique UID Created



Email Verification



LOGIN



Load User's UID



Load User's Private Firestore Data



Farm Configuration



Zones Created



North / South / East / West



Plant Records Created



Plants Grouped Into Sets of 10



Shared Group Addresses



Digital Twin Generated



Manual Sensor Input



Health & Risk Calculation



Digital Twin Updates



Crop Health Updates



Plants Updates



Event Detection



What-If Simulation



Reports



Settings



All Data Saved to Firestore



LOGOUT



Application State Cleared



LOGIN AGAIN

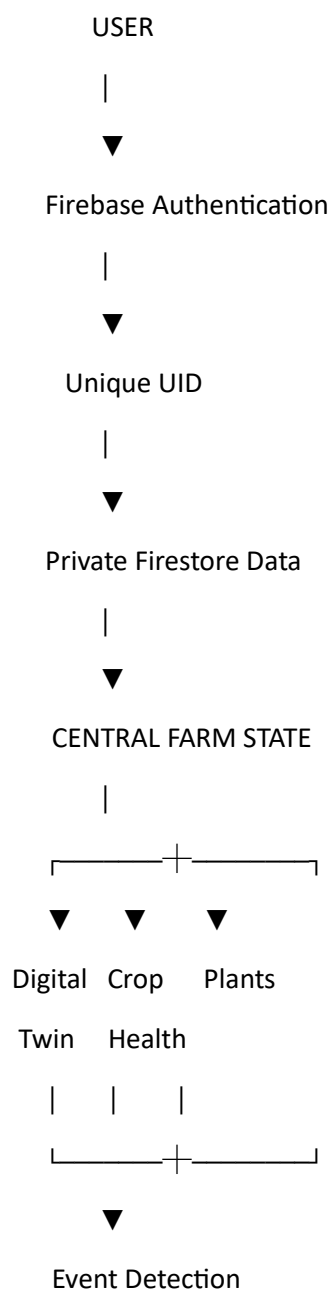


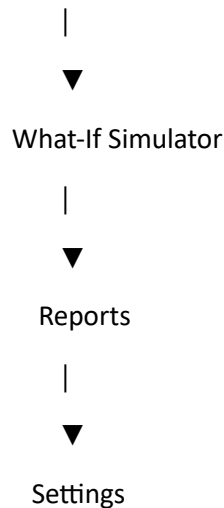
Only That User's Farm Is Restored

33. The Core Idea

AI FarmTwin creates a private digital twin of a user's farm. The user configures the farm, enters environmental values that would normally come from sensors, and the system processes those values to calculate crop health and risk. The results are visualized through a dynamic farm Digital Twin, plant groups, crop-health analytics, event detection, What-If simulation and reports. Firebase Authentication and Firestore ensure that every user has their own completely private and persistent farm environment.

And the overall architecture is:





Manual Testing and Future Hardware Integration

For the current project demonstration, **AI FarmTwin is tested using manually entered sensor values.** Since physical sensors were not connected during the demo, the user can enter values such as **temperature, humidity, soil moisture, and weather conditions** through the website. These values are then processed by the application to calculate the corresponding crop health, stress levels, warnings, and risk conditions. This allows us to demonstrate how the complete digital-twin system responds to changing farm conditions without requiring physical hardware.

How it can work with real hardware in the future

The same system can be extended to a **real IoT-based agricultural monitoring system**. Sensors can be installed in different farm zones and sections, for example:

-  **Temperature sensor** → measures actual temperature
-  **Humidity sensor** → measures air humidity
-  **Soil-moisture sensor** → measures soil water level
-  **Weather sensors** → provide environmental conditions
-  **ESP32/NodeMCU** → collects sensor readings and sends them to the cloud
-  **Firebase/Cloud backend** → stores the sensor data for the authenticated user's farm
-  **AI FarmTwin website** → receives the data and updates the Digital Twin

Demonstration → Real Farm

Current demo:

Manual Sensor Input → Firebase → FarmTwin Processing → Digital Twin → Health/Risk Analysis → Alerts & Reports

Future hardware version:

Physical Sensors → ESP32/IoT Gateway → Firebase/Cloud → FarmTwin → Digital Twin → Health/Risk Analysis → Alerts & Reports

Therefore, **manual sensor input is only the demonstration method used for the current prototype.**

The website architecture is designed so that the manual-input layer can later be replaced or supplemented by real IoT sensors, allowing the Digital Twin to represent **live farm conditions**.

For this prototype, we manually entered sensor readings to demonstrate the complete working of our FarmTwin system because physical hardware was not available during development. The values are processed by the same application logic that would be used with real sensor data. In a real-world deployment, sensors connected through an ESP32 or similar IoT device can automatically send temperature, humidity, soil-moisture and other environmental readings to the cloud. These live readings can then update the farmer's Digital Twin in real time.